

E-Commerce Data Analytics Project



فريق العمل

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Project Documentation Guidelines



Graduate Deadline	Item
1/27/2026	Project Planning & Management
3/1/2026	Literature Review
3/1/2026	Requirements Gathering
4/3/2026	System Analysis & Design
5/17/2026	Implementation (Source Code & Execution)
5/24/2026	Final Presentation & Testing& Reports

Project Planning & Management



At this stage, the work was done as follows:

- 1  **Project Proposal** - This project aims to analyze e-commerce data and transform raw data into actionable insights to support business decision-making and improve overall performance.
- 2  **Objectives** -
 - Analyze customer behavior
 - Measure sales performance (Revenue, Orders, AOV)
 - Analyze payment methods
 - Evaluate customer reviews
 - Identify top products and regions

Project Planning & Management



3



Project Plan -

- **Track 1:** Python + SQL + Power BI dashboard
- **Track 2:** Tableau Dashboard
- **Track 3:** Excel Dashboard

✓ Implemented:

- Data Collection
- Data Cleaning
- Data Analysis
- Database Integration
- Dashboard Development
- Insights & Recommendations

Project Planning & Management



4

 **Roles & Task Assignment** – The project was executed through a collaborative team effort, where all members actively participated in every technical and analytical phase to ensure seamless integration and knowledge sharing:

- **Data Processing (Python)** : The team collaborated on data cleaning, handling missing values, and performing Exploratory Data Analysis (EDA) to ensure high data quality.
- **Database Management (SQL)** : All members contributed to database schema design, writing complex SQL queries, and managing the ETL pipeline to SQL Server.
- **Data Visualization (Power BI / Excel / Tableau)** : Every team member participated in designing and developing interactive dashboards across all three analytical tracks.

Project Planning & Management



5



Risk Assessment -

Risk

Mitigation

Missing Data

Data Cleaning

Duplicate Data

Removing duplicates

Data Quality Issues

Data validation

6



KPIs (Key Performance Indicators) - Project Success Metrics

- **Data Accuracy:** Achieve 100% data integrity during the ETL process from Python to SQL Server.
- **Response Time:** Maintain SQL query execution and dashboard loading time under 3 seconds.
- **System Efficiency:** 100% automation of data cleaning and transformation pipelines.
- **User Adoption:** Multi-platform accessibility through Power BI, Tableau, and Excel dashboards.

Literature Review



1



Importance of Data Analysis

E-commerce data analytics is essential for:

- Improving decision-making
- Increasing revenue
- Enhancing customer experience

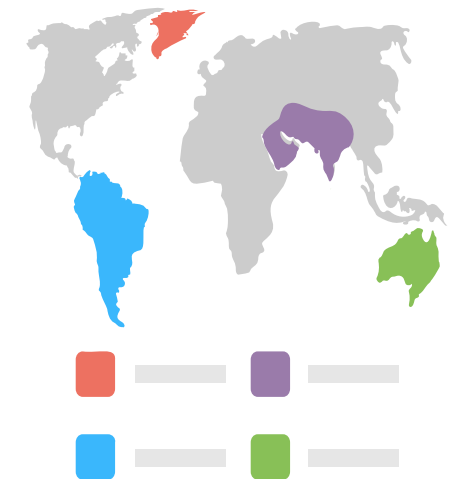


2



Key Concepts Used

- Data Cleaning Techniques
- Exploratory Data Analysis (EDA)
- Business Intelligence Dashboards
- Sales & Customer Analysis



3



Suggested Improvements

- Machine Learning models
- Forecasting techniques
- Recommendation systems

Requirements Gathering



1



Stakeholders

- Management
- Marketing Team
- Operations Team

2



User Stories

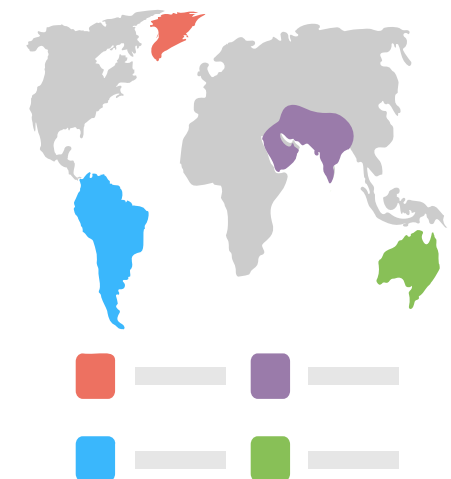
- As a manager, I want to monitor total revenue
- As an analyst, I want to analyze performance trends
- As a user, I want a clear and interactive dashboard

3



Functional Requirements

- Load data from CSV files
- Clean and preprocess data and Merge datasets
- Perform data analysis
- Visualize results in dashboards



Requirements Gathering



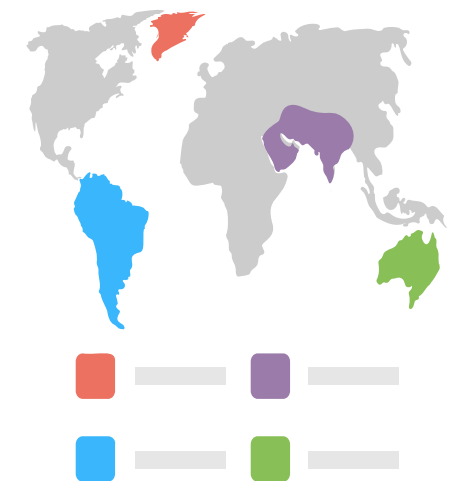
4



Business & Analytical Questions: To guide the analysis and ensure alignment with business objectives, the following key analytical questions were defined:

1. Key Performance Indicators (KPIs)

- What is the Total Revenue generated?
- What is the Total number of unique Orders placed?
- What is the Average Order Value (AOV)?
- What is the Average Shipping Cost (Freight Value) per order?
- What is the Average Delivery Time in days?
- What is the Average Review Score given by customers?
- What are the Total Cancelled Orders and the overall Cancellation Rate %?



2. Time-Based Analysis

- How do Orders change over time (Years, Months)?
- What is the Monthly and Yearly Revenue Trend?

Requirements Gathering



- What time of day (Hour) do customers most frequently place orders?
- How are orders distributed by Day of the Week?
- Is there a clear Seasonal Pattern in sales performance?

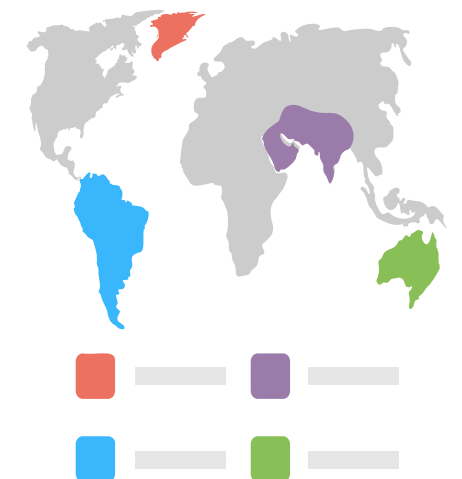
3. Customer & Geographic Analysis

- Which States have the highest number of customers?
- Which Cities generate the most orders?
- Who are the Top 10 Customers based on total payment value?
- What is the ratio of Returning Customers vs. Total Customers?
- How is customer density distributed on a Geographic Concentration Map?



4. Product & Category Analysis

- Which Product Categories generate the highest revenue?
- What are the Top 5 best-selling products by sales value?
- What are the Bottom 5 lowest-selling products?



Requirements Gathering



- Is there a correlation between Product Price and Freight Cost?
- Is there a relationship between Product Weight and Freight Cost?

5. Operations & Logistics Analysis

- What is the distribution of Order Statuses (Delivered, Shipped, Canceled, etc.)?
- What is the Delivery Performance in terms of "On Time" vs. "Late" shipments?
- Which Shipping Methods are used the most and what is their impact on delivery days?
- How are Sellers distributed across different states?



6. Payments & Customer Satisfaction

- What is the revenue distribution by Payment Type (Credit Card, Boleto, Voucher, Debit Card)?
- How are Review Scores distributed (1 to 5 stars)?
- What is the Average Review Score by Order Status?
- What is the total number of Cities served by the platform?

System Analysis & Design



1



Problem Statement : Raw data is unstructured and cannot be directly used for decision-making.

2



System Architecture and Data Flow

CSV Files → Python → SQL Server → Dashboards

Raw Data → Cleaning → Processing → Analysis → Visualization

1. Import Libraries

2. Reading Data

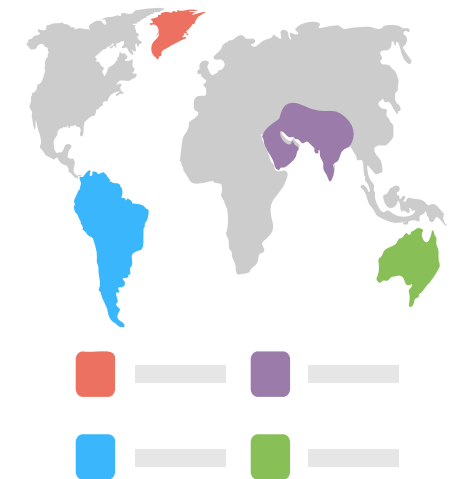
3. Explore Data

4. Data Cleaning

5. Exploratory Data Analysis (EDA)

6. create Dashboard

7. Exporting Data to SQL Database



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Data Cleaning (Example)

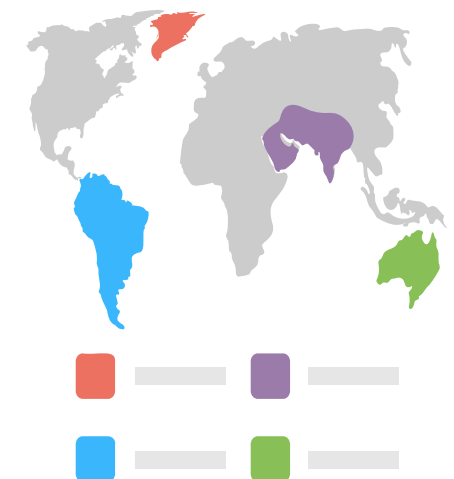
- ◆ Purpose: Convert columns to Date Time format:

```
[11]: date_columns=['order_purchase_timestamp','order_approved_at','order_delivered_carrier_date','order_delivered_customer_da
for col in date_columns:
    Orders[col]=pd.to_datetime(Orders[col])
```

```
[15]: items['shipping_limit_date']=pd.to_datetime(items['shipping_limit_date'])
items.info()
```

- ◆ Purpose: Add anew columns as a month and day name:

```
[13]: Orders['month'] = Orders['order_purchase_timestamp'].dt.to_period('M')
Orders['day_name'] = Orders['order_purchase_timestamp'].dt.day_name()
```



System Analysis & Design



3



Data Cleaning (Example):



Purpose: Handle missing values

```
[12]: Orders[date_columns]=Orders[date_columns].fillna(pd.NaT)
```



4

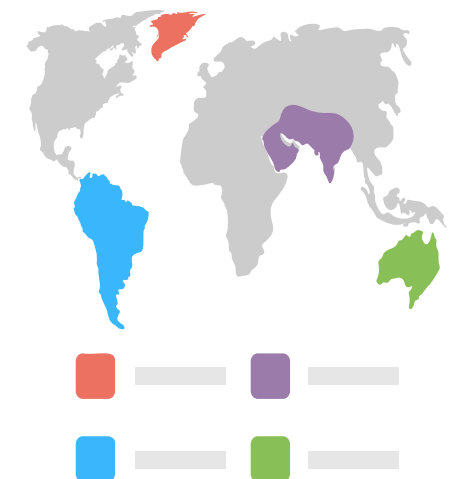


Data Integration:



Purpose: Create a unified dataset

```
[19]: df = Orders.merge(customers,on='customer_id')
df = df.merge(items,on='order_id')
df = df.merge(products,on='product_id')
df = df.merge(payments,on='order_id')
df = df.merge(reviews,on='order_id', how='left')
df.head()
```



System Analysis & Design



5

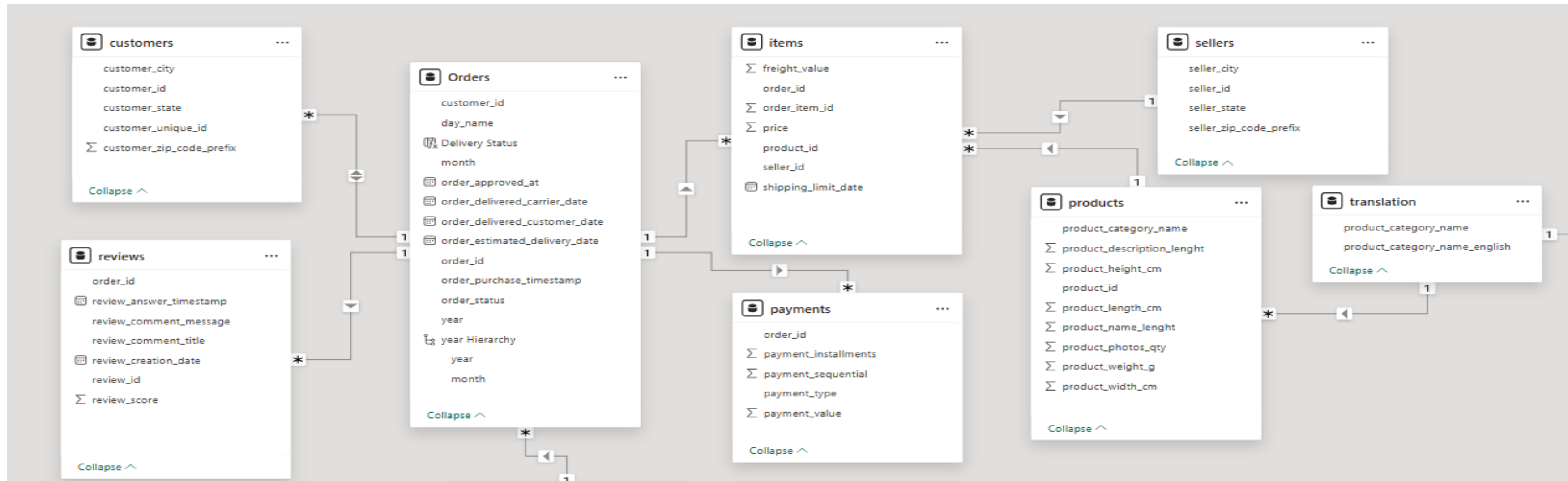
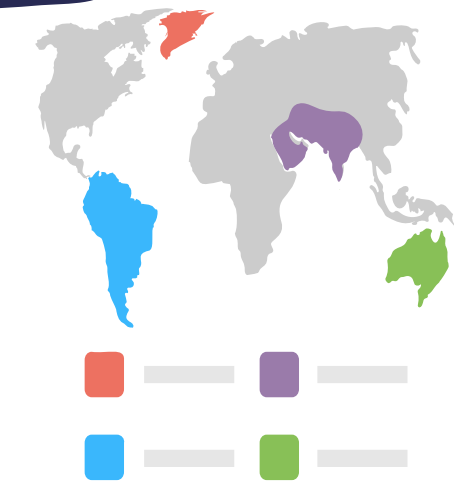


Database Design & Data Modeling

Customers → Orders (1:N)

Orders → Items (1:N)

Orders → Payments (1:N)



Implementation (Source Code & Execution)



✓ Tracks Implementation :

Track I : Python + SQL + Power BI - Implemented:

Data Cleaning > Data Analysis > Database Integration > Dashboard Development

2. Reading Data

```
[6]: customers= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\customers.csv")
Orders= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\Orders.csv")
items= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\order_items.csv")
products= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\products.csv")
payments= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\order_payments_dataset.csv")
sellers= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\sellers.csv")
reviews= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\order_reviews.csv")
geo= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\geolocation.csv")
translation= pd.read_csv(r"E:\مشروع التخرج\4\مشروع الرقمية\راوند\رواد مصر الرقمية\مبادرة\product_category_name_translation.csv")
```

3.Explor Data

```
[8]: dfs = {
    "customers": customers,
    "Orders": Orders,
    "items": items,
    "products": products,
    "payments": payments,
    "sellers": sellers,
    "reviews": reviews,
    "geo": geo,
    "translation": translation
}

def explore_data(df, name="DataFrame"):
    print(f"--- Overview of {name} ---\n")
    print("Shape:", df.shape, "\n")
    print("Sample 5 rows:\n", df.sample(5), "\n")
    print("Columns:", df.columns.tolist(), "\n")
    print("Unique values per column:\n", df.nunique(), "\n")
    print("Info:\n")
    print(df.info(), "\n")
    print("Missing values per column:\n", df.isnull().sum(), "\n")
    print("Duplicate rows:", df.duplicated().sum(), "\n")
    print("-"*50, "\n")

for name, df in dfs.items():
    explore_data(df, name)
```

Implementation (Source Code & Execution)



1



Python Examples :



Question:

Q1. How many total orders do we have?

```
[24]: total_orders = Orders['order_id'].nunique()  
print("total orders:",total_orders)
```

total orders: 99441

Q2: What is the distribution of order status?

```
[26]: order_status = Orders['order_status'].value_counts()  
order_status
```

```
[26]: order_status  
delivered      96478  
shipped        1107  
canceled        625  
unavailable     609  
invoiced        314  
processing      301  
created         5  
approved        2  
Name: count, dtype: int64
```


Implementation (Source Code & Execution)



1



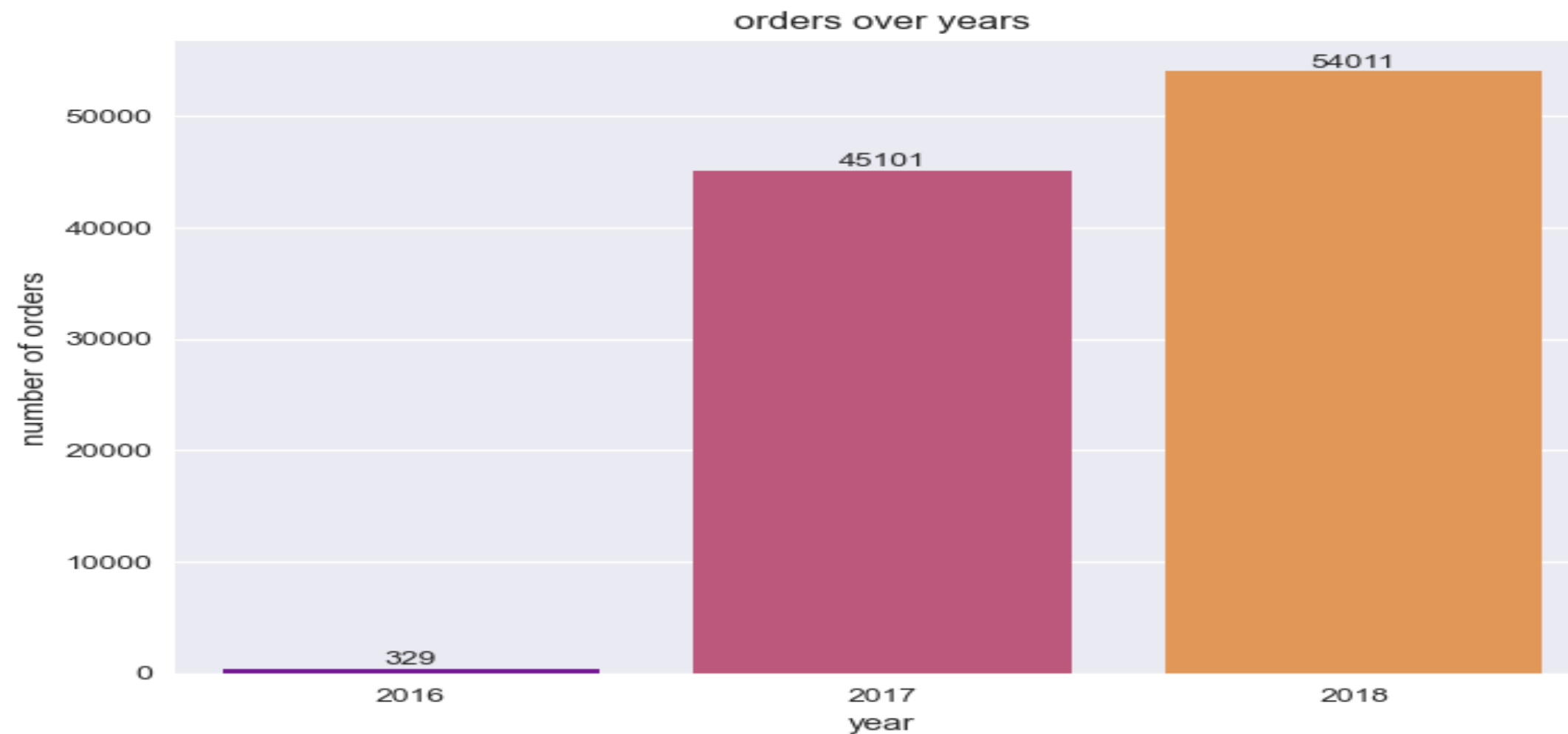
Python Examples :

? Question:

Q3: How do orders change over time?

```
[29]: Orders['year'] = Orders['order_purchase_timestamp'].dt.to_period('y')
      yearly_orders = Orders.groupby('year').size()

      ax=sns.barplot(x=yearly_orders.index.astype(str), y=yearly_orders.values, palette='plasma')
      for i in ax.containers:
          ax.bar_label(i,)
      plt.title("orders over years")
      plt.xlabel("year")
      plt.ylabel("number of orders")
      plt.show()
```



Implementation (Source Code & Execution)



1



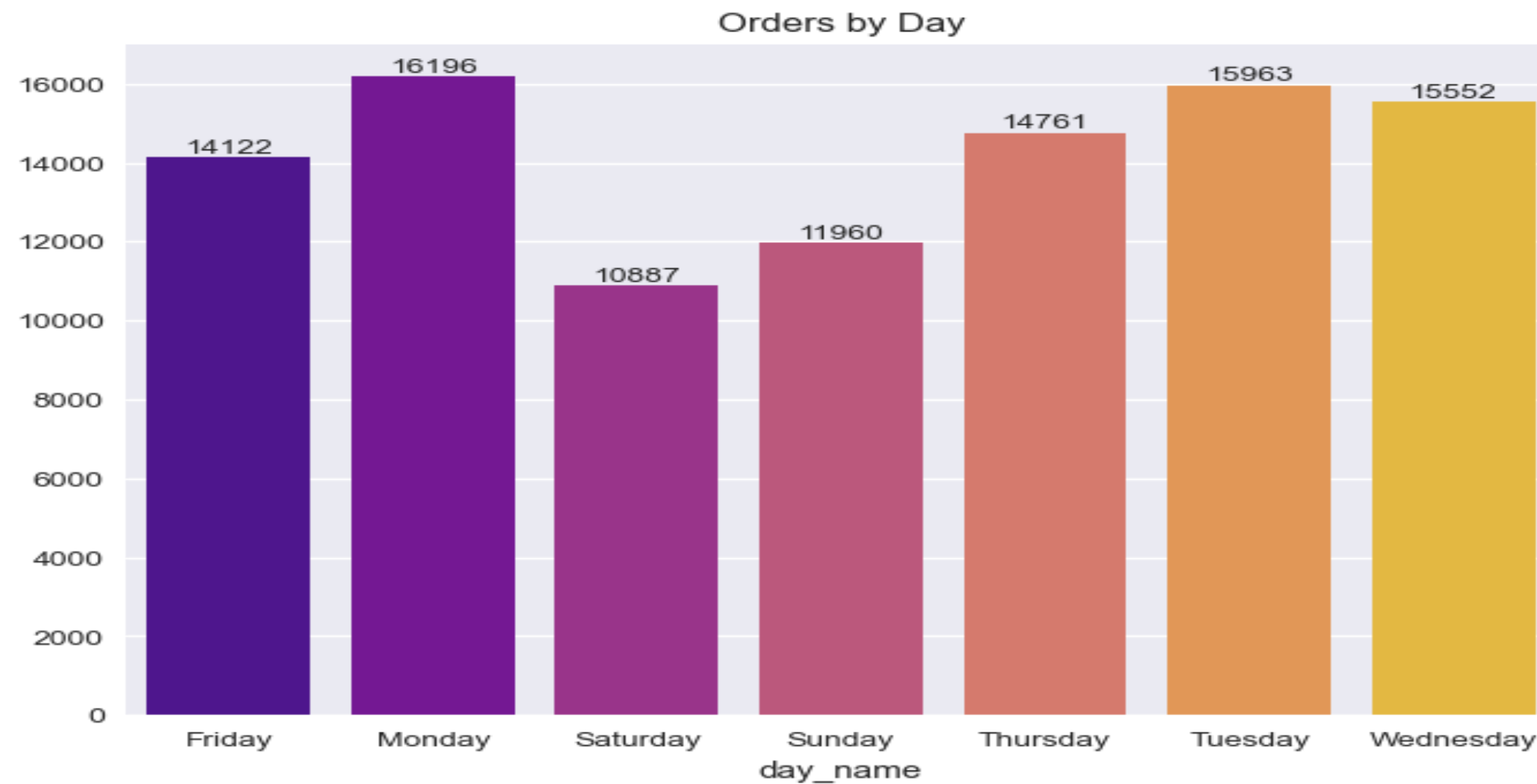
Python Examples :



Question:

```
•[33]: orders_day = Orders.groupby('day_name')['order_id'].count()
ax=sns.barplot(x=orders_day.index,y=orders_day.values,palette='plasma')
for i in ax.containers:
    ax.bar_label(i,)

plt.title("Orders by Day")
plt.show()
```



Implementation (Source Code & Execution)



1



Python Examples :



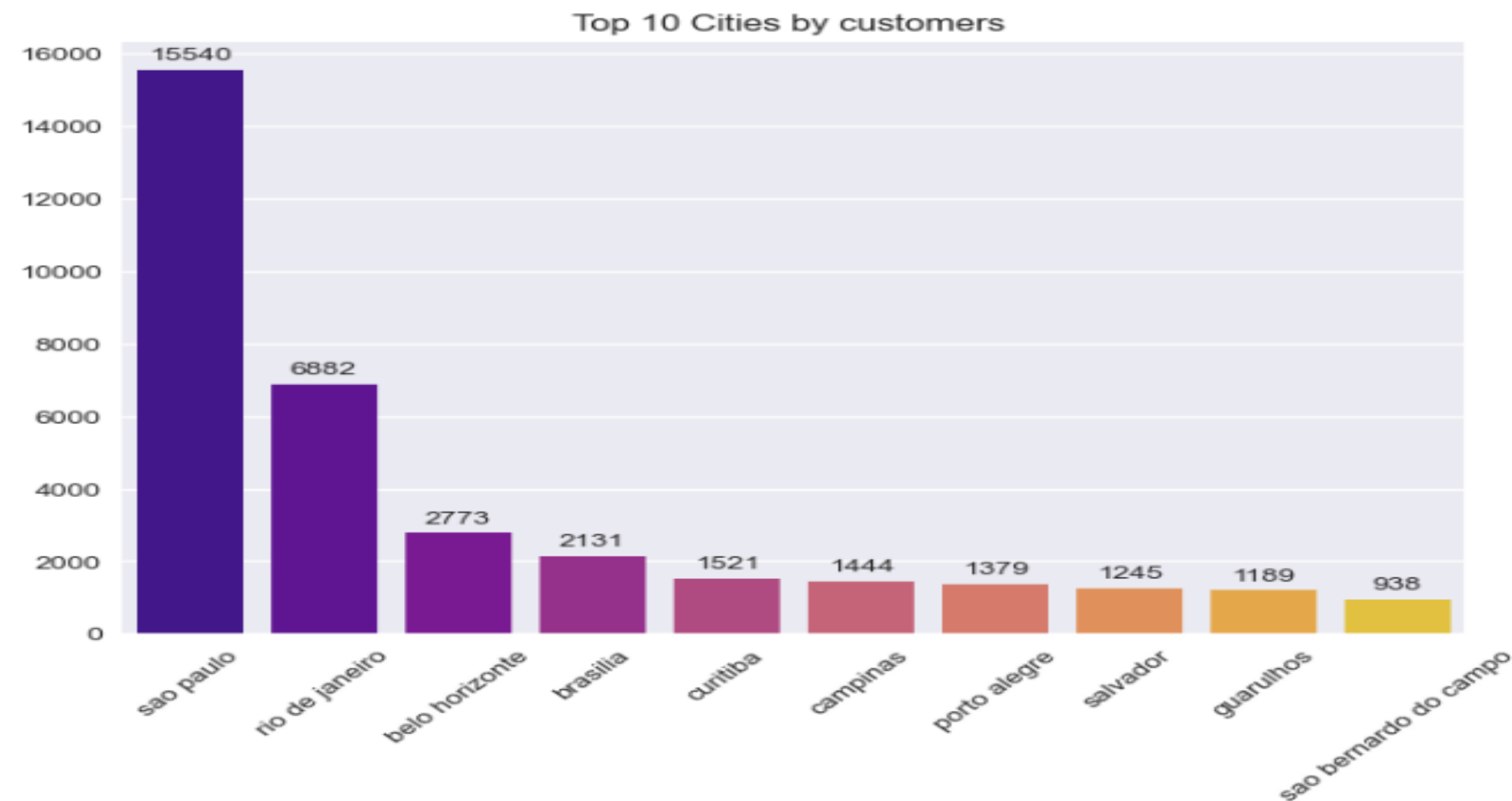
Question:

Q7: Which cities have most customers?

```
[45]: top_cities = customers.groupby('customer_city')['customer_id'].count().sort_values(ascending=False)[:10]

ax = sns.barplot(x=top_cities.index, y=top_cities.values, palette='plasma')
plt.title("Top 10 Cities by customers ")
plt.xticks(rotation=45)

for i in ax.containers:
    ax.bar_label(i, padding=3)
plt.show()
```



Implementation (Source Code & Execution)



2



Python Dashboard :

\$16,008,872

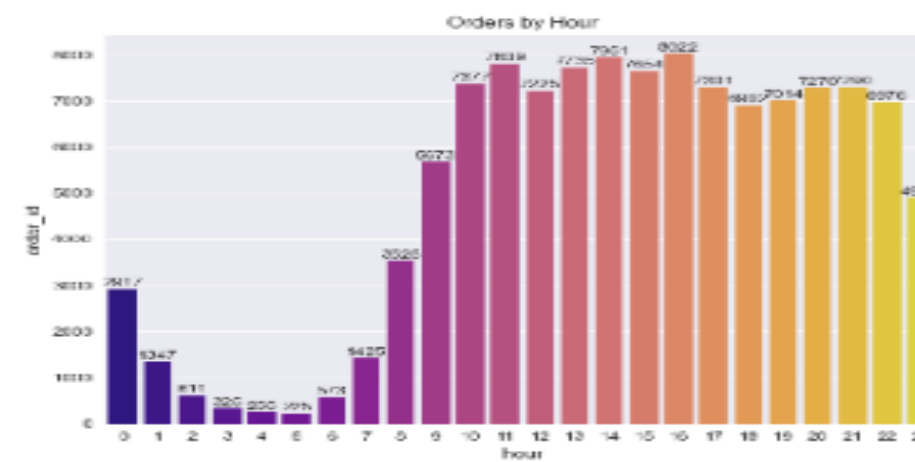
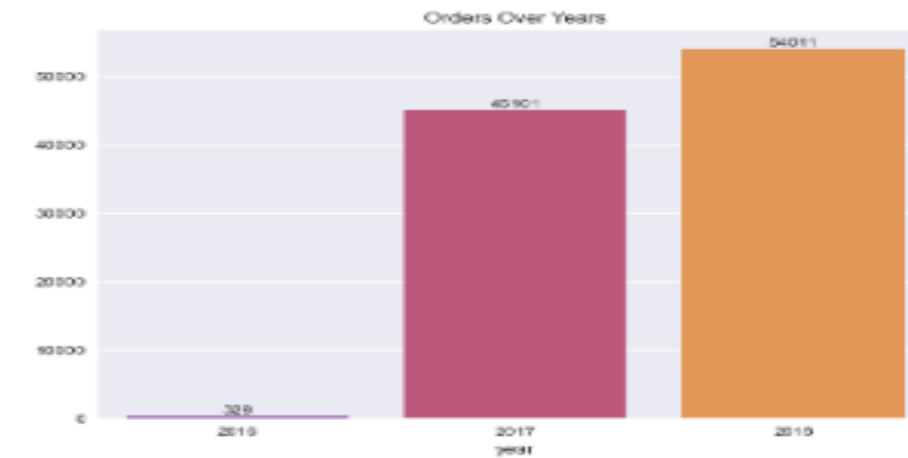
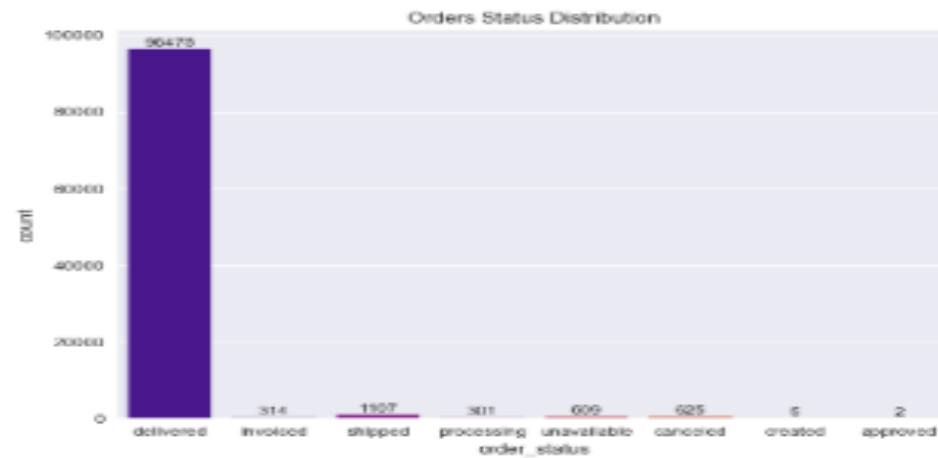
Total_Revenue

99,441

Total Orders

\$161.00

Avg Order Value



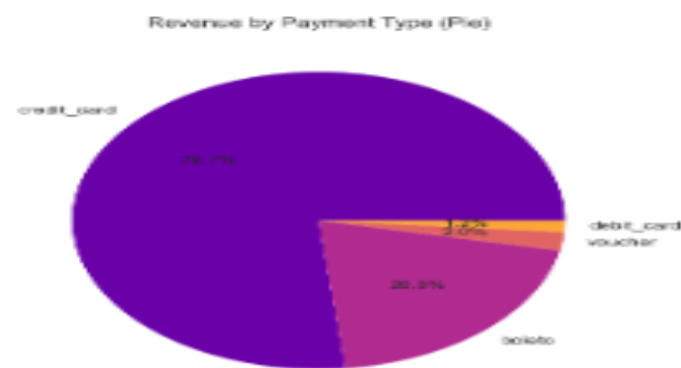
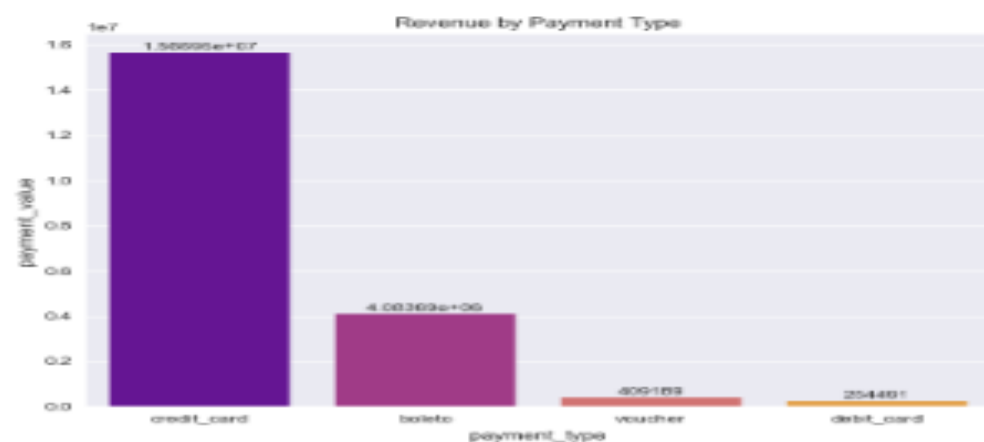
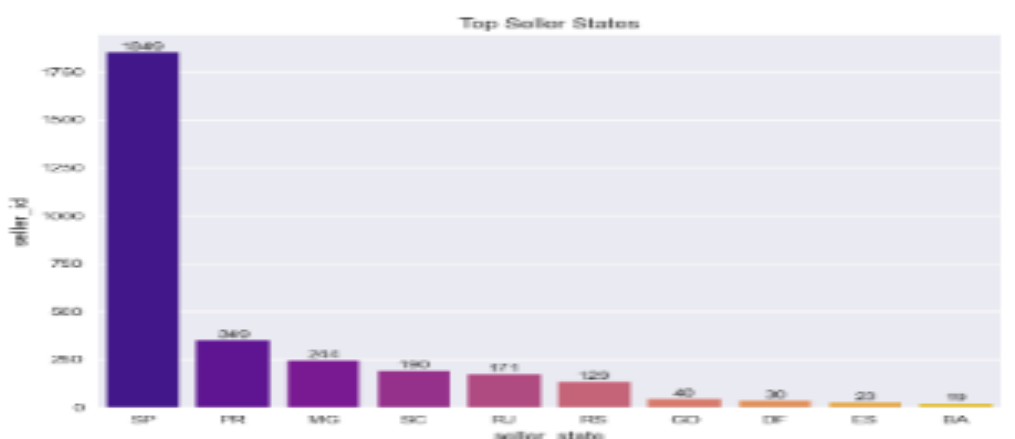
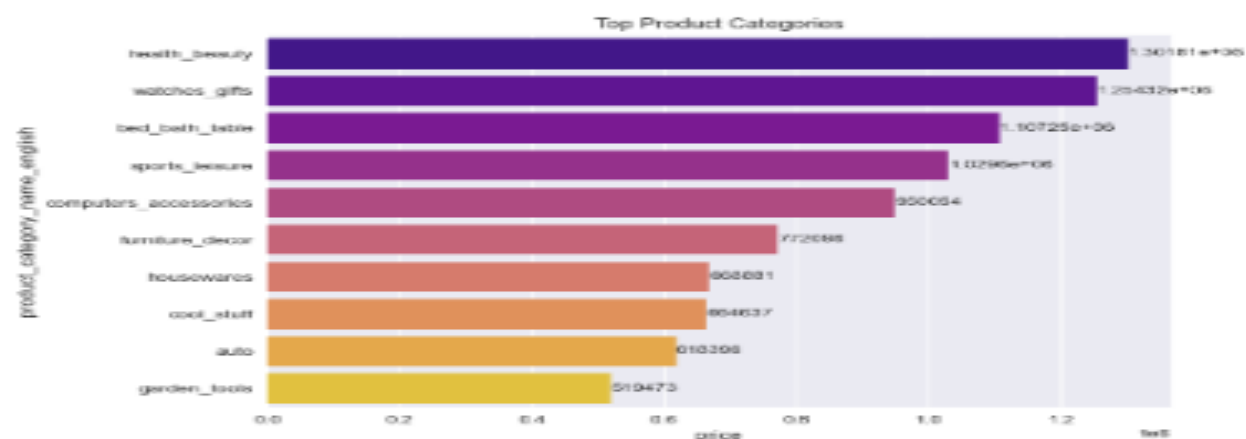
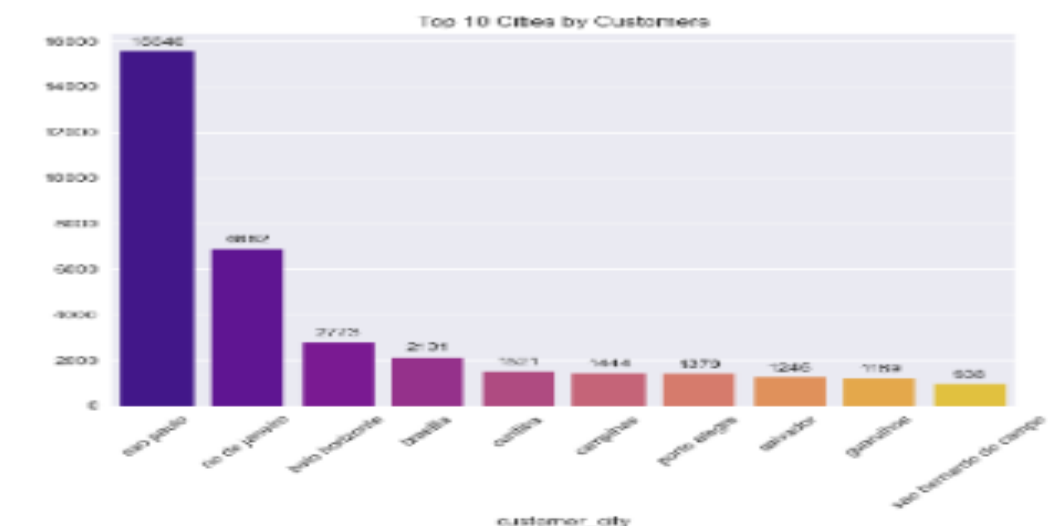
Implementation (Source Code & Execution)



2



Python Dashboard :



Implementation (Source Code & Execution)



3



Exporting Data to SQL Database:

```
[80]: from sqlalchemy import create_engine

SERVER_NAME = "BUSINESS_PLUS"
DATABASE_NAME = "E-commerce_Analytics"

connection_string = (
    f"mssql+pyodbc://{SERVER_NAME}/{DATABASE_NAME}"
    "?driver=ODBC+Driver+17+for+SQL+Server"
    "&trusted_connection=yes"
)
engine = create_engine(connection_string)
print("✅ !يتاح SQL Server تم الاتصال ب")

for col in Orders.columns:
    if str(Orders[col].dtype).startswith("period"):
        Orders[col] = Orders[col].astype(str)
        print(f"String إلى Period من '{col}' تم تحويل عمود")

tables = {
    "customers" : customers,
    "Orders" : Orders,
    "items" : items,
    "products" : products,
    "payments" : payments,
    "sellers" : sellers,
    "reviews" : reviews,
    "geo" : geo,
    "translation" : translation,
}

for table_name, df in tables.items():
    print(f"بمرفق جدول: {table_name} ...")

    df.to_sql(
        name = table_name,
        con = engine,
        if_exists = "replace",
        index = False,
    )
```

Implementation (Source Code & Execution)



4



SQL Examples

? Question: What is the total revenue?:

? Question: What is the average order value?

```
select  
round(SUM(payment_value),0) as total_paymens,  
round(AVG (payment_value),0) as average_order_value  
from payments
```

Results			Messages	
	total_paymens	average_order_value		
1	16008872	154		

? Question: payment methods by unique order count and total value?

```
select  
payment_type,  
count( order_id)as order_count,  
round(SUM(payment_value),0) as total_paymens  
from payments  
group by payment_type  
order by total_paymens desc
```

Results				Messages	
	payment_type	order_count	total_paymens		
1	credit_card	76795	12542084		
2	boleto	19784	2869361		
3	voucher	5775	379437		
4	debit_card	1529	217990		
5	not_defined	3	0		

Implementation (Source Code & Execution)



4



SQL Examples

--average review score for orders

```
select
order_status,
avg(review_score) as average_score,
count(review_id) as total_reviews
from Orders o
join reviews r on o.order_id=r.order_id
group by order_status
order by average_score desc
```

	order_status	average_score	total_reviews
1	delivered	4	96361
2	approved	2	2
3	shipped	2	1043
4	created	2	3
5	unavailable	1	597
6	canceled	1	609
7	invoiced	1	313
8	processing	1	296

--orders with payments higher than 4000

```
select
order_id,
sum(payment_value) as total_order_value
from payments
group by order_id
having sum(payment_value) > 4000
order by total_order_value desc
```

	order_id	total_order_value
1	03caa2c082116e1d31e67e9ae3700499	13664.08
2	736e1922ae60d0d6a89247b851902527	7274.88
3	0812eb902a67711a1cb742b3cdaa65ae	6929.31
4	fefacc66af859508bf1a7934eab1e97f	6922.21
5	f5136e38d1a14a4dbd87dff67da82701	6726.66
6	2cc9089445046817a7539d90805e6e5a	6081.54
7	a96610ab360d42a2e5335a3998b4718a	4950.34
8	b4c4b76c642808cbe472a32b86cddc95	4809.44
9	199af31afc78c699f0dbf71fb178d4d4	4764.34
10	8dbc85d1447242f3b127dda390d56e19	4681.78
11	426a9742b533fc6fed17d1fd6d143d7e	4513.32
12	d2f270487125ddc41fd134c4003ad1d7	4445.5
13	80dfedb6d17bf23539beef3c768f4d7	4194.76
14	68101694e5c5dc7330c91e1bbc36214f	4175.26
15	b239ca7cd485940b31882363b52e6674	4163.51
16	9a3966c23190dbdbaabed08e8429c006	4042.74

Implementation (Source Code & Execution)



5

Power BI Dashboard :



Implementation (Source Code & Execution)



5

Power BI Dashboard :



Implementation (Source Code & Execution)



5 Power BI Dashboard :



Implementation (Source Code & Execution)



5 Power BI Dashboard :



 Question: Key Performance Indicators (KPIs)

- | | | | | | | |
|--------------|------------------|---------------|-----------------|-------------------|-------------------|------------|
| Total Orders | Cancelled Orders | Total Revenue | Avg Order Value | Avg Shipping Cost | Avg Delivery Days | Avg Review |
| 99.44K | 625 | \$16.01M | \$161 | \$20 | 12.50 | 4 |

Cancellation Rate %
0.63%

Implementation (Source Code & Execution)

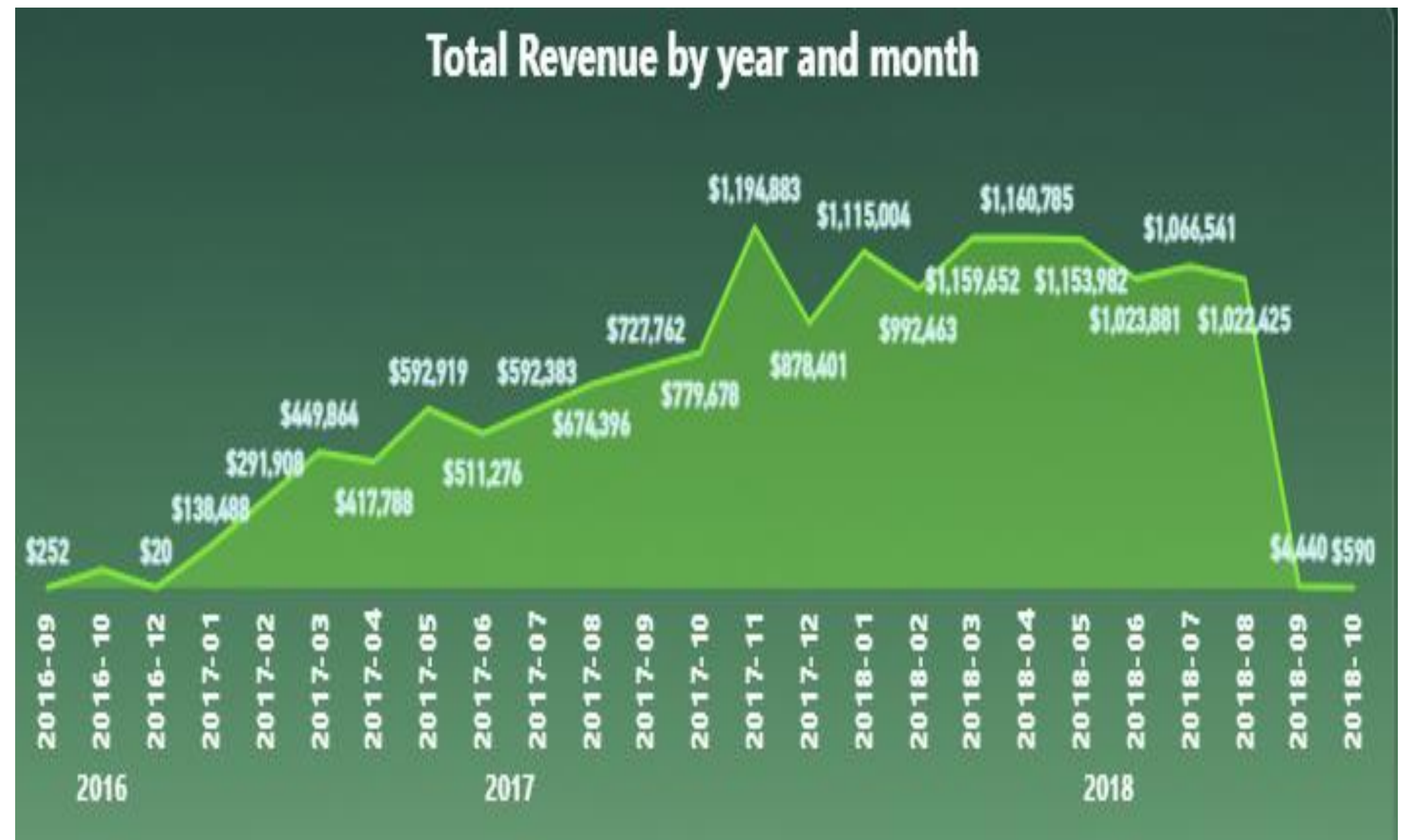
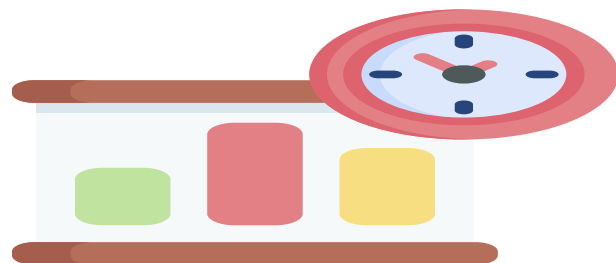


Power BI Examples :



Question: What is the Monthly and Yearly Revenue Trend?

Revenue shows a strong upward trend starting from early 2017, reaching its all-time peak of \$1.19M in November 2017. The business maintained a stable monthly performance above \$1M throughout the first half of 2018.



Implementation (Source Code & Execution)



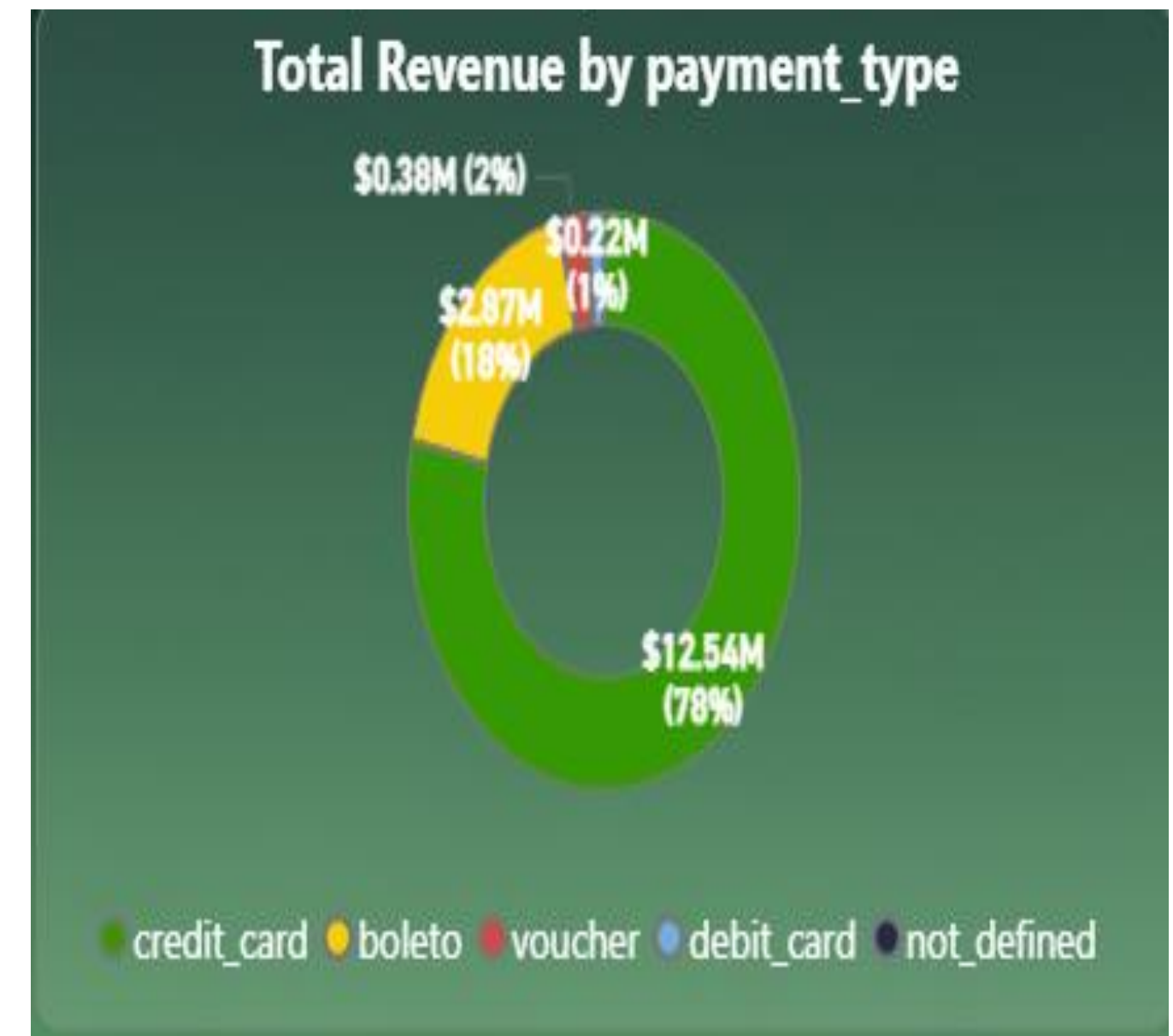
Power BI Examples :



Question: • What is the revenue distribution by Payment Type?

The Credit Card is the dominant payment method, accounting for 78% (\$12.54M) of total revenue, followed by Boleto at 18% (\$2.87M). Other methods like Voucher and Debit Card contribute minimally, representing only 3% of the total revenue combined.

Insights :High Credit Card usage (78%) indicates a preference for installment payments, while the Boleto share (18%) emphasizes the need for cash-based options to ensure full market coverage.



Implementation (Source Code & Execution)



Power BI Examples :



Question: • What is the distribution of Order Statuses ?

The vast majority of orders have been successfully Delivered (96,478), while other statuses like Shipped (1,107), Canceled (625), and Unavailable (609) represent only a tiny fraction of the total volume.

Insights : The high volume of delivered orders (over 97%) indicates a very high fulfillment success rate and operational efficiency. The low number of canceled or unavailable orders suggests effective inventory management and a reliable supply chain.



Implementation (Source Code & Execution)



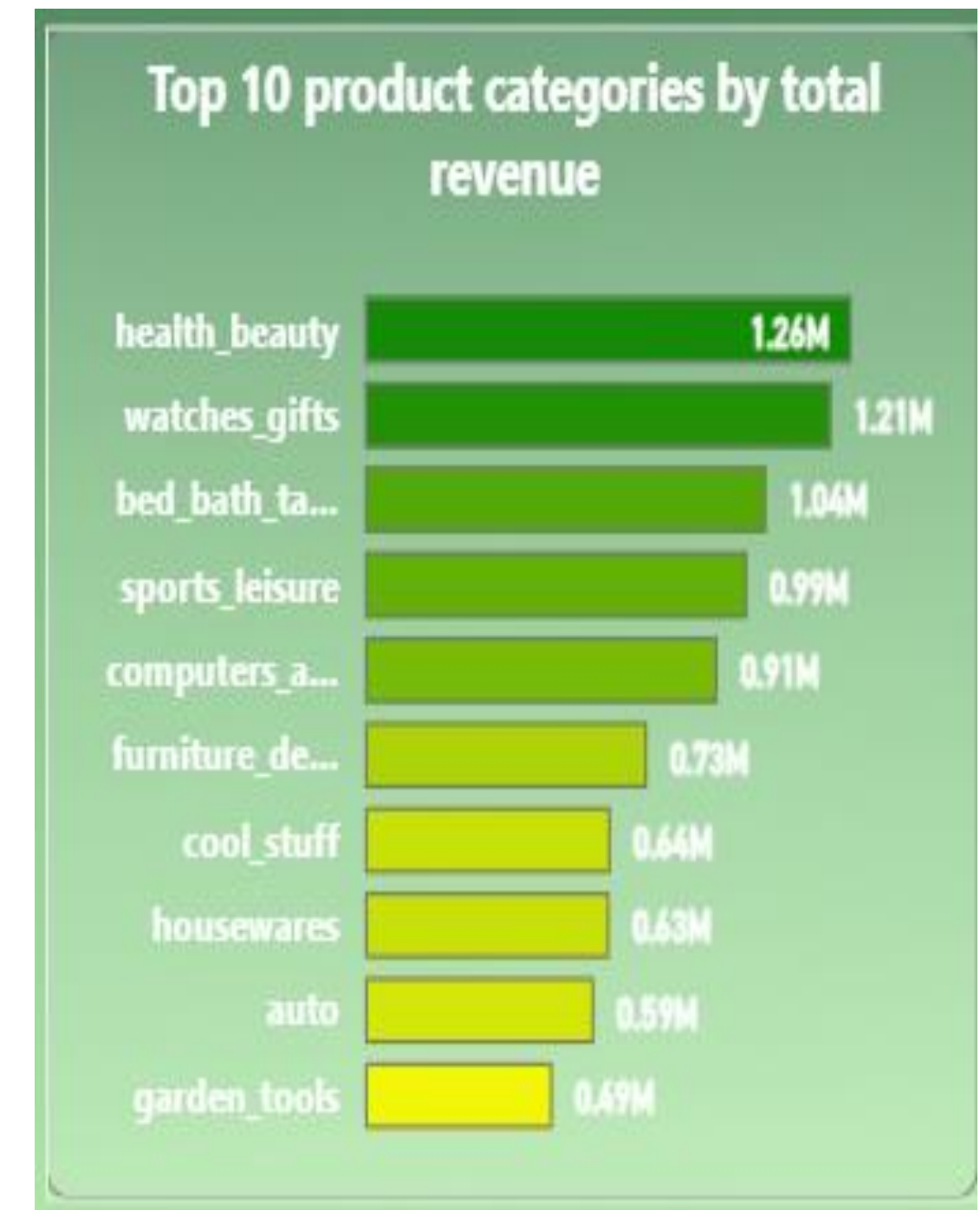
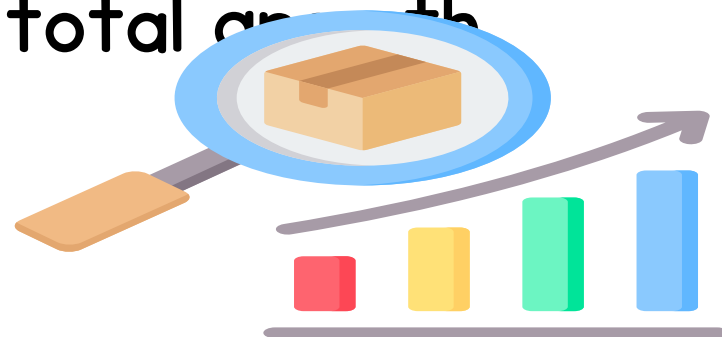
Power BI Examples :



Question: · Which Product Categories generate the highest revenue?

The Health & Beauty category generates the highest revenue at \$1.26M, followed closely by Watches & Gifts at \$1.21M and Bed, Bath, & Table at \$1.04M.

Insights : Lifestyle and personal care products dominate the platform's revenue, suggesting a customer base that prioritizes wellness and gifting .The small revenue gap between the top categories indicates a diversified market where multiple sectors contribute significantly to total growth.



Implementation (Source Code & Execution)



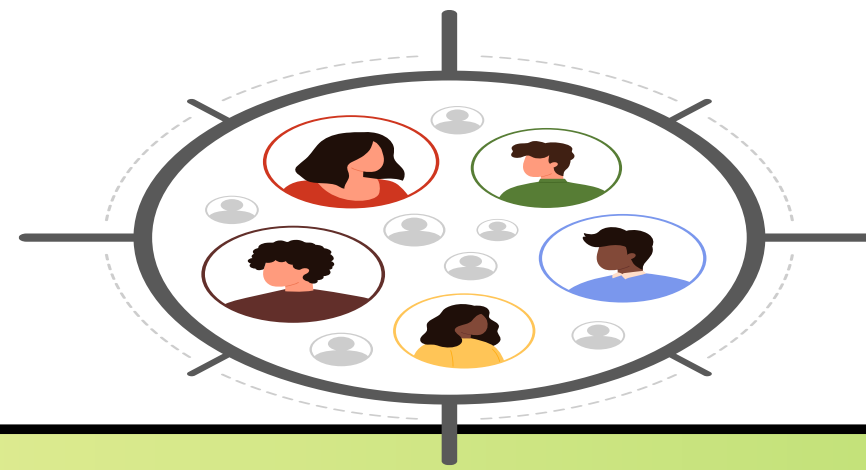
Power BI Examples :



Question: • Which City have the highest number of customers?

Sao Paulo has the highest number of customers by a significant margin, with 15.54K customers.

Insights: Sao Paulo is the primary hub for the business, containing more than double the customers of the second-ranking city, Rio de Janeiro. The concentration of customers in large urban centers like Sao Paulo and Rio de Janeiro suggests that marketing and logistics efforts are most effective in major metropolitan areas.



Total Customers by customer_city

sao paulo	15.54K
rio de janeiro	6.88K
belo horizonte	2.77K
brasilia	2.13K
curitiba	1.52K
campinas	1.44K
porto alegre	1.38K
salvador	1.25K
guarulhos	1.19K
sao bernardo...	0.94K

Implementation (Source Code & Execution)



Power BI Examples :



- Question:
- How are Review Scores distributed (1 to 5 stars)?
 - What is the Average Review Score by Order Status?



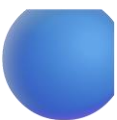
Average Review Score by Status: The "delivered" status holds the highest average review score of 4, while statuses like "shipped," "canceled," and "unavailable" drop to an average of 2, and "processing" has the lowest average at 1. Overall Distribution: The total average review score for all 99,224 reviews is 4, indicating generally high customer satisfaction for completed orders.

Insights: There is a clear correlation between order completion and customer satisfaction, as "delivered" orders significantly outperform all other statuses in review scores. Low scores for "processing" and "shipped" statuses suggest that delays or uncertainties during the fulfillment phase negatively impact customer perception even before delivery.

Review Score Analysis by Order Status

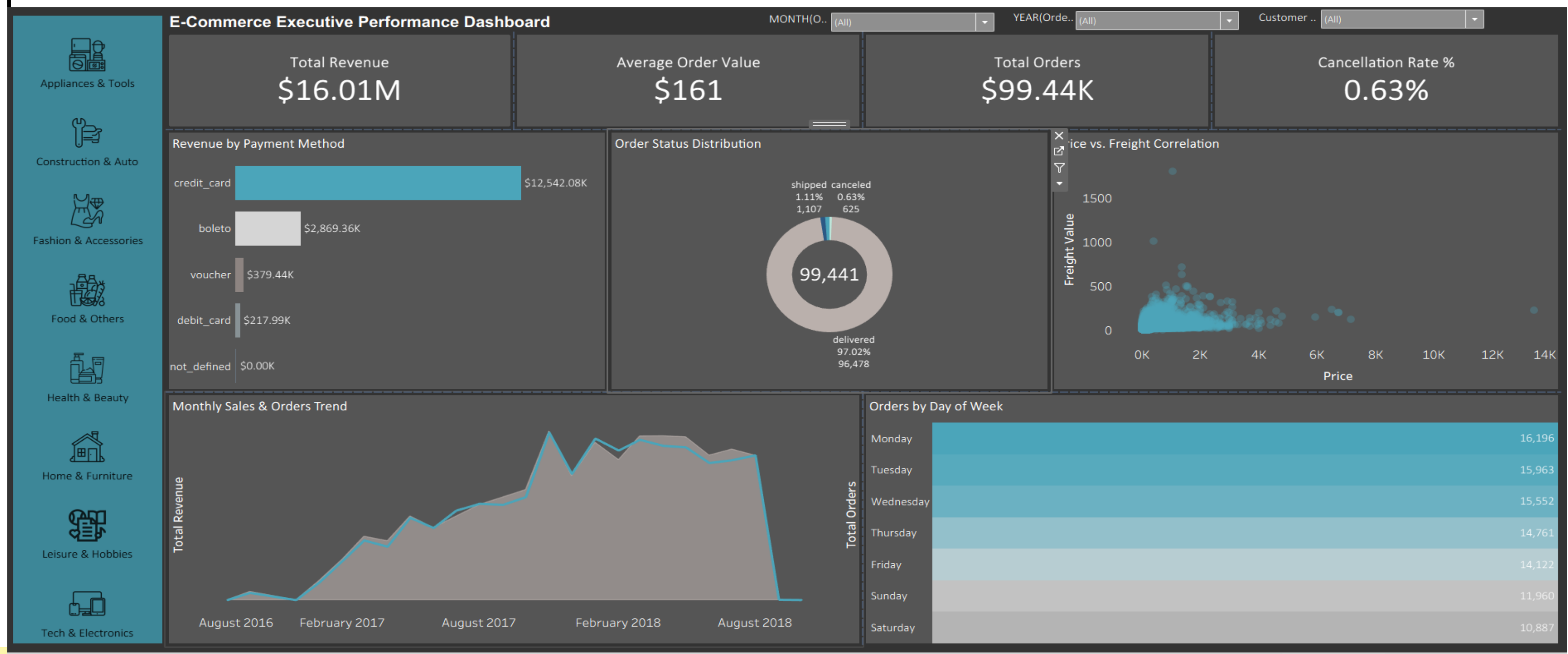
order_status	Total Orders	Count of review_id	Avg Review
delivered	96478	96361	4
shipped	1107	1043	2
canceled	625	609	2
unavailable	609	597	2
invoiced	314	313	2
processing	301	296	1
created	5	3	2
approved	2	2	3
Total	99441	99224	4

Implementation (Source Code & Execution)



Track 2 : Tableau Dashboard - Implemented:

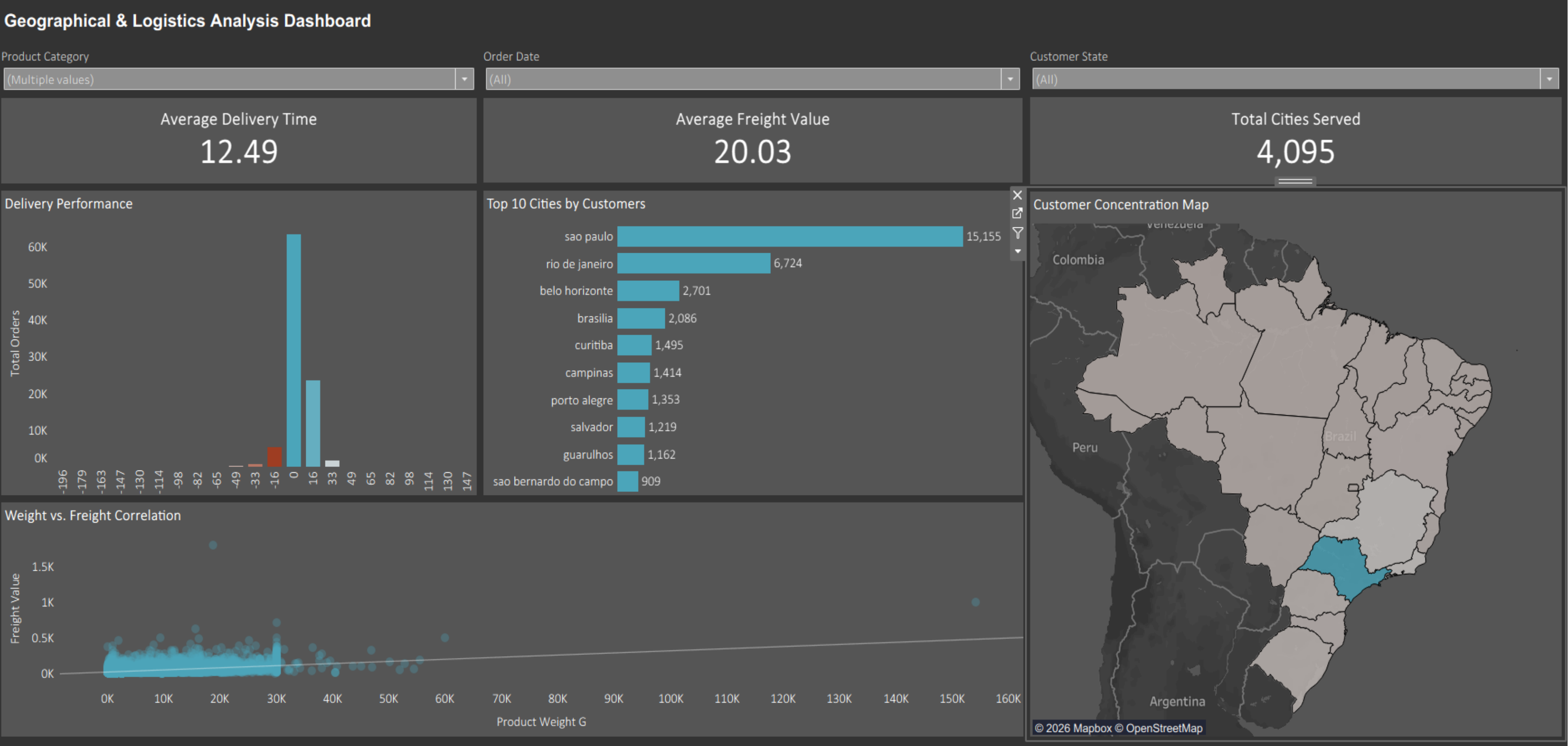
Data Cleaning > Data Analysis Data > Integration > Dashboard Development



Implementation (Source Code & Execution)



Tableau Dashboard -



Implementation (Source Code & Execution)



Tableau Examples :

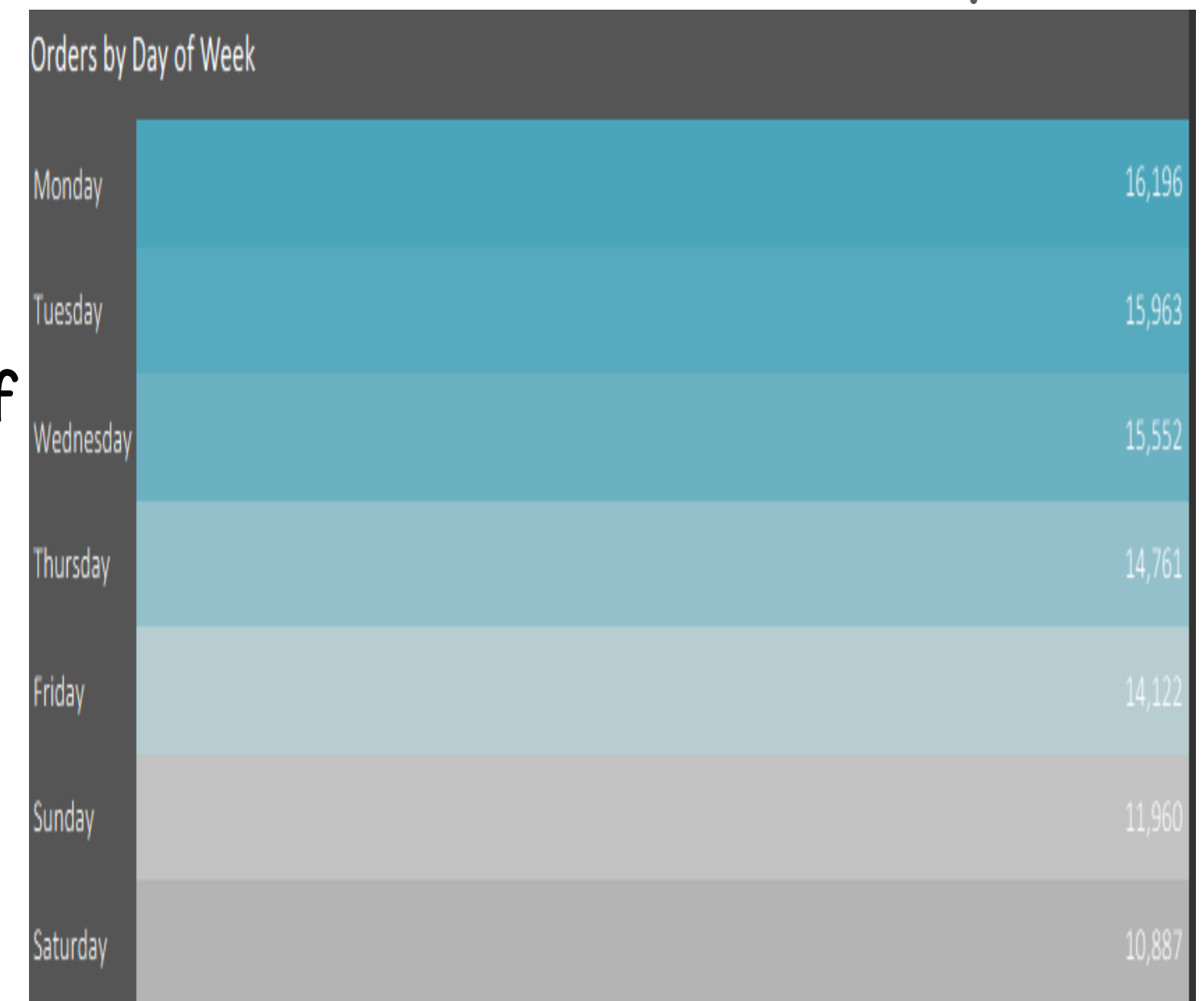
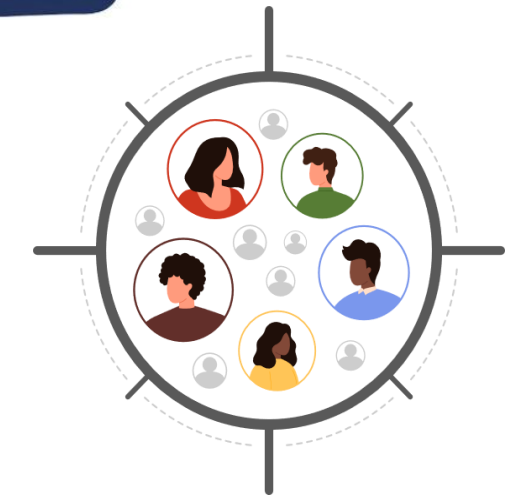


Question: • How are orders distributed by Day of the Week?

Order volume peaks at the beginning of the week, with Monday recording the highest number of orders at 16,196, followed by Tuesday and Wednesday. Activity then gradually declines, reaching its lowest point on Saturday with 10,887 orders.

Insights: The customers prefer shopping at the start of the work week to ensure delivery before the next weekend.

The drop in activity during the weekend (Saturday and Sunday) indicates that promotional campaigns or "flash sales" would be most effective if launched late Sunday or early Monday to capture the peak demand period.



Implementation (Source Code & Execution)



Track 3 : Excel Dashboard – Implemented:

Data Cleaning > Data Analysis > Data Integration > Dashboard Development

Final Presentation & Testing& Reports



◆ Final Insights Summary -

Market Dominance: Sao Paulo stands as the primary economic hub, contributing significantly to the total customer base of 99,441.

Financial Growth: Revenue achieved an explosive upward trend from 2017 to mid-2018, peaking at \$1.19M in November 2017.

Consumer Preference: Credit Cards are the undisputed payment choice (78%), and Health & Beauty is the top-performing category by revenue (\$1.26M).

Operational Excellence: The platform maintains an exceptional fulfillment rate with 96,478 successfully delivered orders and an overall average review score of 4/5.

Correlation Between Fulfillment and Rating: The significant gap between the average review score of delivered orders (4/5) versus processing orders (1/5) highlights that customer satisfaction is heavily tied to delivery speed and communication transparency. This suggests that technical or logistical bottlenecks during the initial order phases have a disproportionately negative impact on the brand's reputation, regardless of the final product quality.

Final Presentation & Testing& Reports



◆ Recommendations :

Logistics Optimization: Prioritize expanding warehouse capacity and distribution networks in high-density regions like Sao Paulo and Rio de Janeiro to further reduce delivery times.

Payment Strategy: While credit cards lead, enhancing the Boleto and Debit Card user experience could capture underserved market segments that prefer cash-based or immediate payments.

Marketing Timing: Align "Flash Sale" campaigns and promotional emails with Monday and Tuesday mornings to capitalize on the weekly peak in order volume.

Category Expansion: Replicate the successful marketing and supply chain strategies of the Health & Beauty and Watches categories in lower-performing sectors like Auto and Garden Tools to balance revenue streams.

Customer Retention: Implement a proactive feedback loop for "Processing" and "Shipped" statuses to address the lower review scores observed during the fulfillment phase.

Final Presentation & Testing& Reports



◆ Recommendations :

Dynamic Pricing & Promotional Strategy: Use the identified peak periods (Mondays and Tuesdays) to implement dynamic pricing or limited-time offers to maximize conversion rates when customer intent is at its highest.

Customer Segmentation & Targeted Marketing: Develop a Loyalty Program specifically for customers in high-density cities like Sao Paulo and Rio de Janeiro to increase "Customer Lifetime Value" (CLV) and reduce acquisition costs.

Basket Analysis & Cross-Selling: Leverage the success of the Health & Beauty and Watches categories by implementing a "Frequently Bought Together" recommendation engine to increase the average order value (AOV).

Boleto Payment Incentives: Offer small "Instant Discounts" for customers using Boleto or Debit Cards to encourage immediate payment and improve the company's cash flow liquidity compared to long-term credit card installments.